11th International Conference in Vietnam on the Development of Biomedical Engineering
BME 11 | HEALTHCARE 4.0: SMART HEALTH, PERSONALIZE MEDICINE AND ARTIFICIAL INTELLIGENCE FOR LOW-AND MIDDLE- INCOME COUNTRIES
Cam Ranh - Nha Trang, Vietnam
July, 2026

Circulating hsa-miR-23b-3p as a Candidate Biomarker for Pan-Cancer Detection Across 13 Cancer Types

Thi Tuong Van Nguyen[1], Huu Tran Don Phu[1], Thi Kim Nhuong Nguyen[2], Thi Thu Hien Pham[3,*] Minh Nam Nguyen[4,*]

[1]Faculty of Biology and Biotechnology, University of Science, Vietnam National University Ho Chi Minh City (VNU-HCM), Ho Chi Minh City, Vietnam

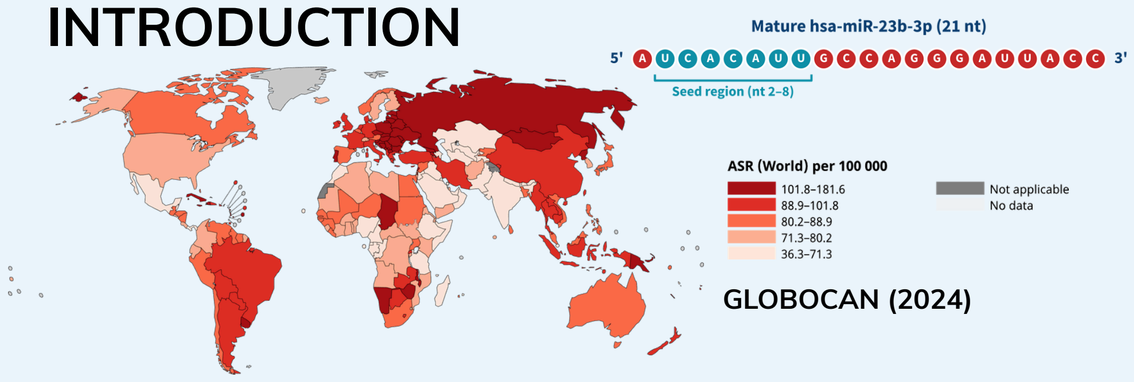
[2]Research Center for Genetics and Reproductive Health Vietnam National University Ho Chi Minh City, University of Health Sciences, VNU-HCM, Ho Chi Minh City, Vietnam

[3] School of Biomedical Engineering, International University, Vietnam National University Ho Chi Minh City (VNU-HCM), Ho Chi Minh City, Vietnam

[4]Department of Biomedical Engineering, Faculty of Medicine, University of Health Sciences, VNU-HCM, Ho Chi Minh City, Vietnam

* Corresponding author: nmnam@uhsvnu.edu.vn; ptthien@ hcmiu.edu.vn

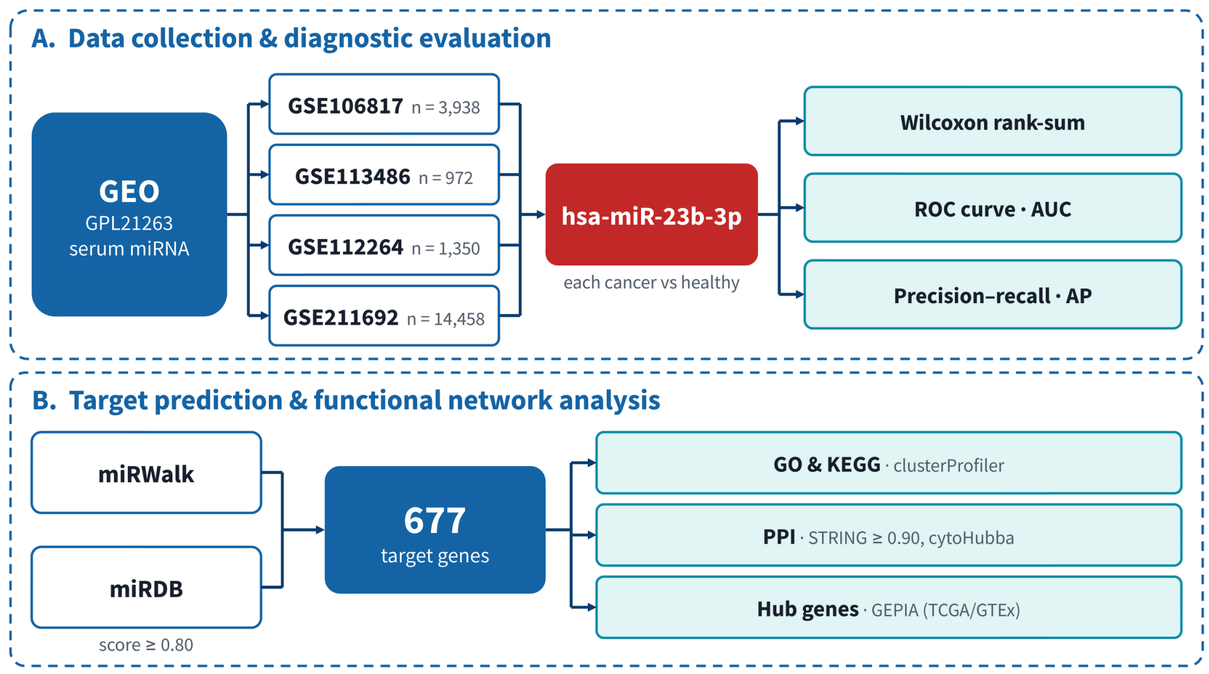
INTRODUCTION



13 cancer types | **4** GEO cohorts | **20 718** serum samples | **1** single miRNA

AIM: Evaluate whether circulating hsa-miR-23b-3p alone can discriminate cancer patients from healthy controls across 13 cancer types, and explore its biological relevance through target-gene and network analysis.

METHODOLOGY



RESULTS & DISCUSSION

1 Elevated expression in all 13 cancers

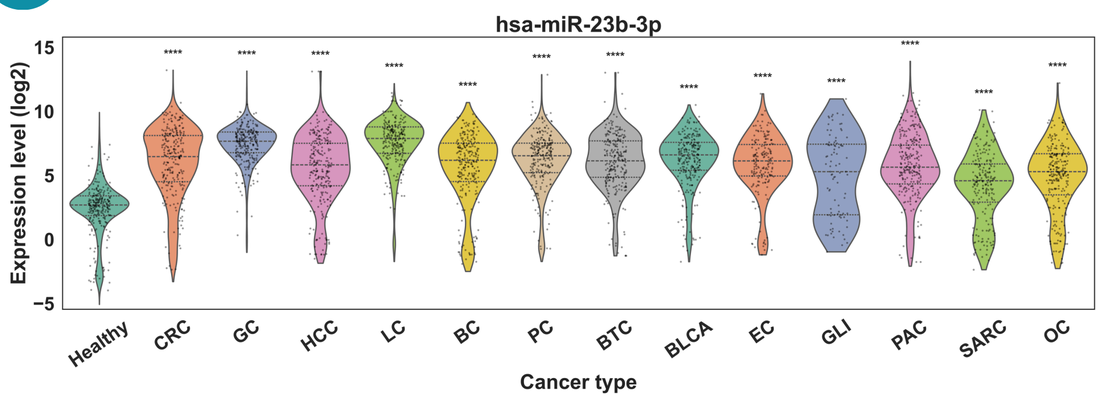


Fig. 1. Expression of hsa-miR-23b-3p in healthy controls and 13 cancer types. All cancers differ from healthy controls (Wilcoxon rank-sum, **** p < 0.0001).

2 Evaluation of hsa-miR-23b-3p discrimination

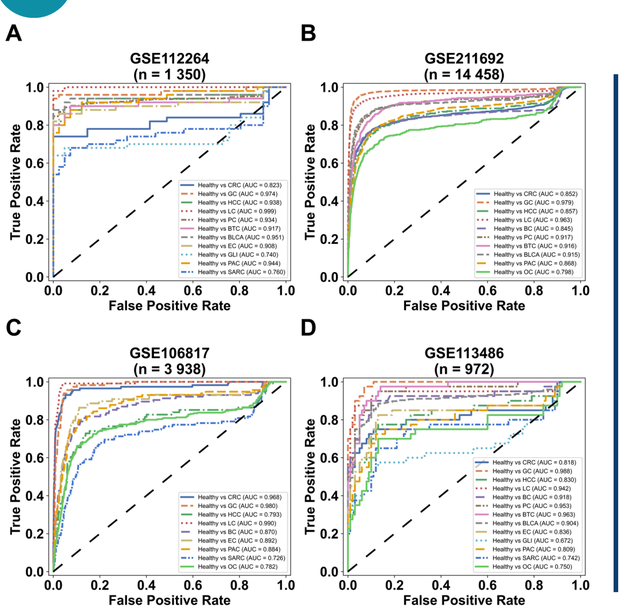


Fig. 2. ROC curves of hsa-miR-23b-3p for each cancer type vs healthy controls in four GEO cohorts: (A) GSE112264 (n = 1,350), (B) GSE211692 (n = 14,458), (C) GSE106817 (n = 3,938), (D) GSE113486 (n = 972)

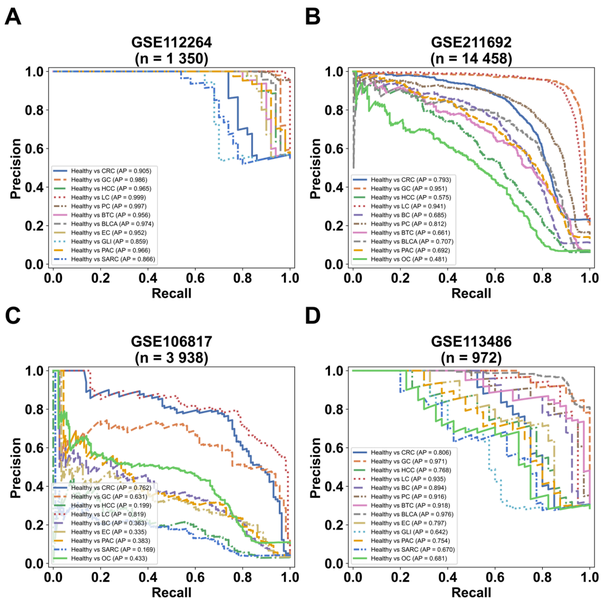


Fig. 3. PR curves of hsa-miR-23b-3p for each cancer type vs healthy controls in four GEO cohorts: (A) GSE112264 (n = 1,350), (B) GSE211692 (n = 14,458), (C) GSE106817 (n = 3,938), (D) GSE113486 (n = 972)

3 Performance across four cohorts

	GSE112264 (n = 1,350)		GSE113486 (n = 972)		GSE106817 (n = 3,938)		GSE211692 (n = 14,458)	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP
Lung (LC)	0.999	0.999	0.942	0.935	0.990	0.819	0.963	0.941
Gastric (GC)	0.974	0.986	0.988	0.971	0.980	0.631	0.979	0.951
Biliary tract (BTC)	0.917	0.956	0.963	0.918	–	–	0.916	0.661
Prostate (PC)	0.934	0.997	0.953	0.916	–	–	0.917	0.812
Bladder (BLCA)	0.951	0.974	0.904	0.976	–	–	0.915	0.707
Esophageal (EC)	0.908	0.952	0.836	0.797	0.892	0.335	–	–
Breast (BC)	–	–	0.918	0.894	0.870	0.363	0.845	0.685
Pancreatic (PAC)	0.944	0.966	0.809	0.754	0.884	0.383	0.868	0.692
Colorectal (CRC)	0.823	0.905	0.818	0.806	0.968	0.762	0.852	0.793
Hepatocellular (HCC)	0.938	0.965	0.830	0.768	0.793	0.199	0.857	0.575
Ovarian (OC)	–	–	0.750	0.681	0.782	0.433	0.798	0.481
Sarcoma (SARC)	0.760	0.866	0.742	0.670	0.726	0.169	–	–
Glioma (GLI)	0.740	0.859	0.672	0.642	–	–	–	–

Score (AUC / AP)

Fig. 4. Area under the ROC curve (AUC) and average precision (AP) for hsa-miR-23b-3p in each cancer type across the four GEO datasets

4 Enrichment of hsa-miR-23b-3p Target Genes

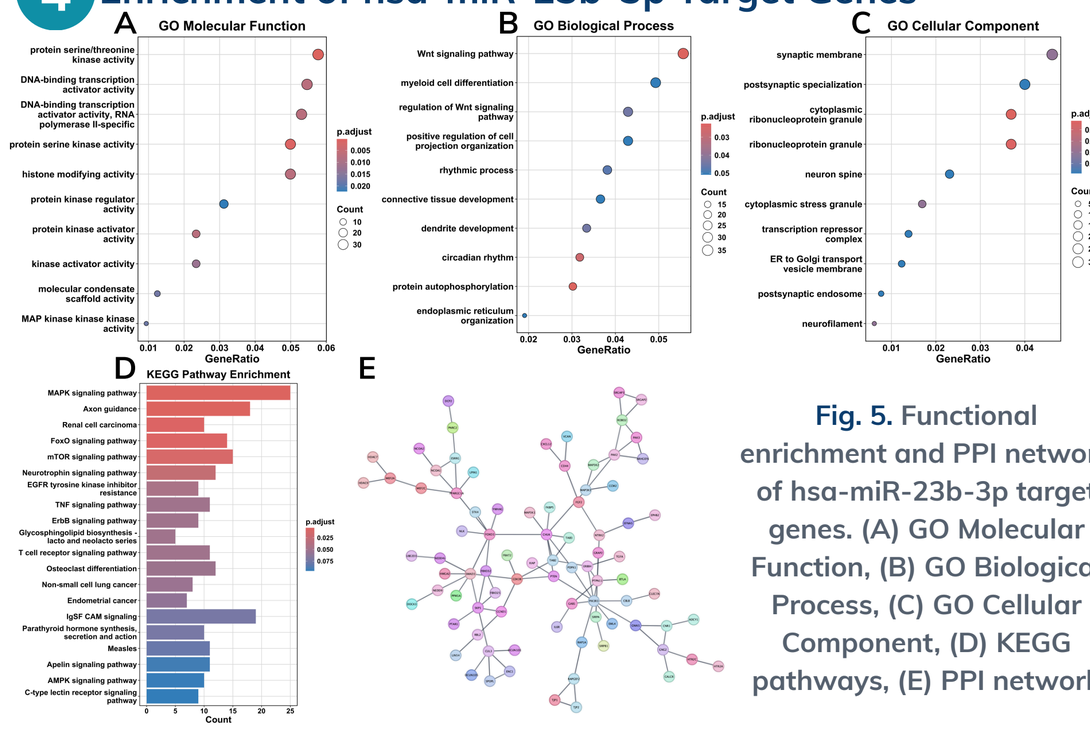


Fig. 5. Functional enrichment and PPI network of hsa-miR-23b-3p target genes. (A) GO Molecular Function, (B) GO Biological Process, (C) GO Cellular Component, (D) KEGG pathways, (E) PPI network

CONCLUSION

- Circulating hsa-miR-23b-3p alone distinguished cancer patients from healthy controls across 13 cancer types with moderate-to-high accuracy.
- It performed best in lung, gastric and pancreatic cancers, and this pattern held across four independent cohorts.
- Its predicted target genes were enriched in cancer-related pathways (MAPK, mTOR, FoxO, ErbB), which still needs experimental validation.
- hsa-miR-23b-3p is a promising candidate for minimally invasive first-line screening, but prospective, multi-platform validation is needed before clinical use.